Crime Reporting Dataset

Coverage: January 2024 - December 2025 • Forces: Dorset, Gloucestershire, Warwickshire and Wiltshire

Reporting grain: police force × month × crime category × location status × LSOA (where available).

Output: 171,406 aggregated reporting rows representing 436,692 retained incidents.

1 STATEMENT OF WORK

1.1.1 Project Objective

Develop a reliable, reporting ready dataset to support crime analysis across Dorset, Wiltshire, Gloucestershire and Warwickshire police forces, covering January 2024 to December 2025.

The dataset should enable stakeholders to investigate:

- Crime trends, category distributions, seasonal patterns and unusual spikes.
- Comparisons between police forces including population adjusted crime rates.
- Relationships between crime, population and socioeconomic deprivation.
- Recurring seasonal patterns.
- Geographical crime hotspots and differences between police forces (optional).

1.1.2 Scope of Work

- Ingest, clean and validate UK Police crime data using Python.
- Investigate missing values, duplicates and inconsistent records.
- Aggregate crime data by police force, month, crime category and geographical area.
- Enrich the dataset with population, income deprivation and geographical information.
- Validate the final dataset for accuracy, completeness and reporting suitability.

1.1.3 Deliverables

- Reproducible Jupyter Notebook containing the data engineering pipeline.
- Cleaned aggregated and enriched CSV dataset.

- Data dictionary describing the final dataset.
- Workflow diagram documenting the pipeline.

1.1.4 Assumptions and Limitations

Crime locations are anonymised, population and deprivation datasets use different reference years. These limitations will be documented and considered when interpreting the data.

1.1.5 Completion Criteria

The pipeline runs successfully from raw data to export, validation checks pass and the final dataset is suitable for use in Power BI.

2 DATA DICTIONARY

Field	Type	Definition	Source and derivation	Usage / limitations
police_force	Text	Reporting police force	Police.uk street crime CSV (Reported by); standardised. Raw force name retained/standardised for joining	Dimension; force year population repeats across categories and locations
year	Integer	Calendar year of the reporting month	Derived from Police.uk Month. First four characters of month converted to integer	Used to join annual force population not a separate observation
month	Text (YYYY-MM)	Crime reporting month	Police.uk Month. Retained as YYYY-MM	Convert to date in Power BI if required for proper time axis
crime_category	Text	Crime classification supplied in crime data	Police.uk Crime type. Aggregated by source crime category	ASB is reported separately from recorded crime
incident_type	Text	Distinguishes antisocial behaviour from recorded crimes	Derived from Crime type. ASB for Anti-social behavior, Recorded crime otherwise	Categorisation follows the source crime type
location_status	Text	Whether a record has a usable English LSOA or an alternative location classification	Derived from LSOA codes and validation checks. Records with missing, Welsh or conflicting location data retained in separate status groups	Review exact category labels in export; status groups should not be mapped as English LSOAs
lsoa_code	Text	2021 Lower Layer Super Output Area code for valid English locations	Police.uk LSOA code, matched against English deprivation geography. Retained only for matched English LSOAs	Blank for nonmatched locations; use for geographical joins
lsoa_name	Text	Official source LSOA name	Police.uk LSOA name. Retained at matched LSOA grain	Blank when no matched English LSOA

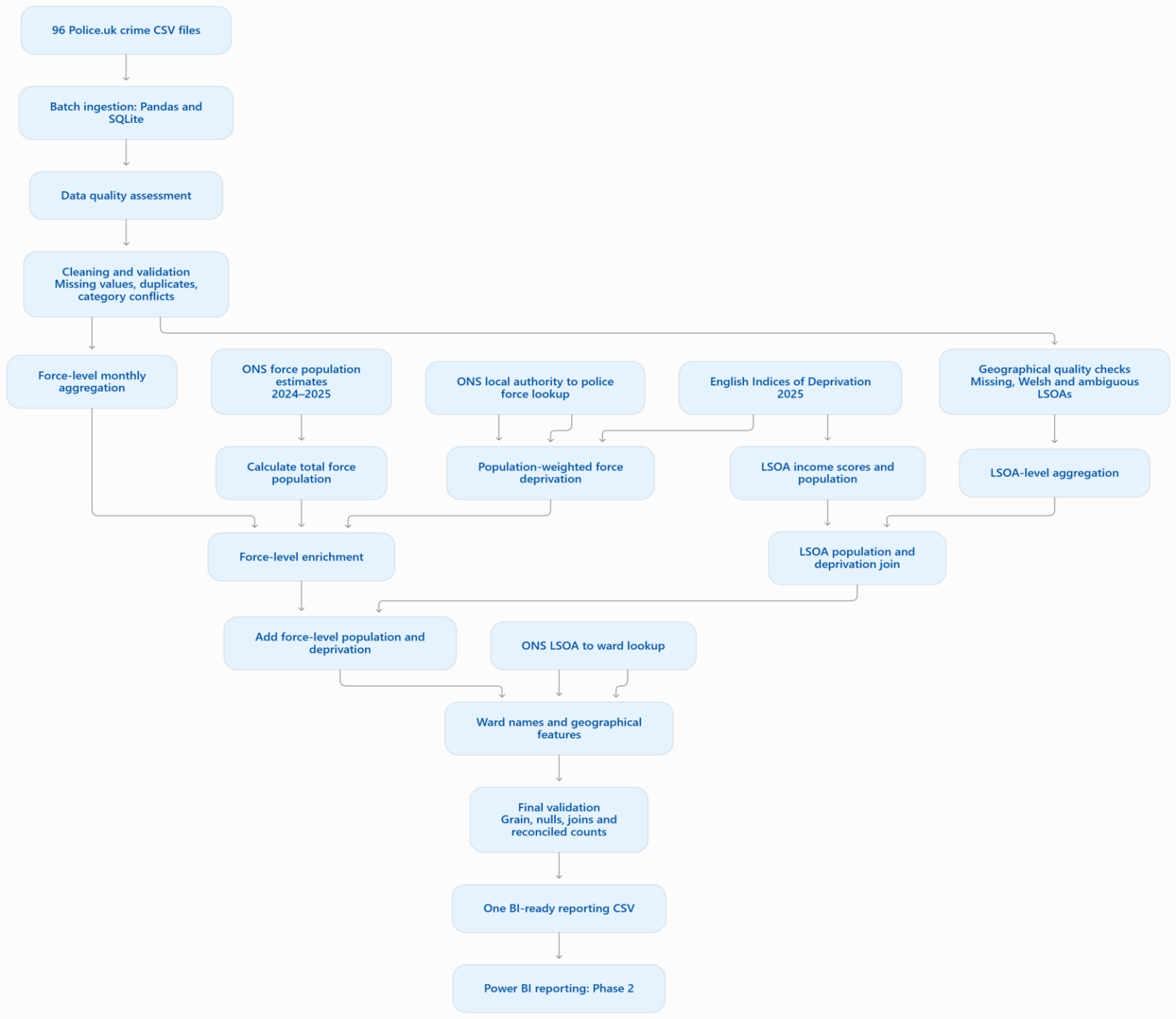
Field	Type	Definition	Source and derivation	Usage / limitations
incident_count	Integer	Number of incidents in this reporting row	Cleaned Police.uk crime records. Group size at force × month × crime category × location status × LSOA (where valid)	Additive across mutually exclusive groups; dataset total equals cleaned incident count
lsoa_population_2022	Decimal	2022 population denominator for the assigned LSOA	English Indices of Deprivation data: Total population mid 2022. Matched by LSOA code; repeated across category/month rows	Not additive across reporting rows; 2022 denominator used for 2024–25 comparisons
lsoa_income_deprivation_rate	Decimal (0–1)	LSOA income deprivation score/rate	English Indices of Deprivation 2025: Income Score (rate). Matched by LSOA code	Contextual socioeconomic measure, not a monthly measure and not necessarily causal
lsoa_monthly_rate_per_1000	Decimal	Monthly incidents for category and LSOA per 1,000 residents	Calculated from incident_count and 2022 LSOA population. incident_count / lsoa_population_2022 × 1,000	Calculate rates again after grouping, do not sum precomputed rates
force_population	Integer	Annual mid year population of police force area	ONS police-force-area mid-year estimates, 2024–25. Female and male age bands summed; joined by police force and year	Repeated across LSOAs/categories/months; do not sum across reporting rows
force_income_deprivation_rate	Decimal (0–1)	Population weighted income deprivation rate for force area	English Indices of Deprivation 2025 and 2022 LSOA denominators. Σ(LSOA income rate × LSOA population) / Σ(LSOA population), aggregated to force	Constant contextual measure across 2024–25, not an observed annual trend
ward_name	Text	Electoral ward name assigned to the LSOA	ONS 2021 LSOA to 2022 ward best fit lookup (WD22NM). Many-to-one best-fit match on LSOA code	Best fit assignment, an LSOA is not necessarily wholly within its assigned ward
local_authority	Text	Local authority component of the source LSOA name	Derived from LSOA name. Parsed from text preceding final numbered LSOA identifier	Check extraction for nonstandard source names
lsoa_identifier	Text	Short alphanumeric suffix of the LSOA name (e.g. 035B)	Derived from LSOA name. Regular expression extraction of three digits plus capital letter at end	Not unique nationally, always join using lsoa_code

3 ASSUMPTIONS AND LIMITATIONS

- Geographical completeness: unmatched English LSOA enrichment is intentionally missing for Welsh, unknown or ambiguous locations. Such incidents remain included in force level counts.
- Population reference periods: force level denominators use matching 2024/2025 annual estimates; local LSOA denominators use mid 2022 population and are fixed across both years.
- Ward mapping: electoral ward names use a 2022 best fit mapping from 2021 LSOAs and do not guarantee exact boundary overlap.

- Deprivation: 2025 income deprivation scores provide contextual comparisons for both crime years not annual changes or evidence of causality.
- Population fields repeat across months and categories, do not sum them directly in Power BI. Calculate rates from appropriately aggregated incident counts and population denominators.
- Antisocial behaviour is identified separately from recorded crime. Identical ASB rows without unique identifiers are retained; identifiable crime records were deduplicated by force, month and Crime ID.
- Crime locations are approximate/anonymised. The reporting dataset aggregates to LSOA instead of preserving individual incident coordinates.

4 WORKFLOW DIAGRAM


96 Police.uk crime CSV files
Batch ingestion: Pandas and SQLite
Data quality assessment
Cleaning and validation
Missing values, duplicates, category conflicts
Force-level monthly aggregation
ONS force population estimates 2024–2025
ONS local authority to police force lookup
English Indices of Deprivation 2025
Geographical quality checks
Missing, Welsh and ambiguous LSOAs
Calculate total force population
Population-weighted force deprivation
LSOA income scores and population
LSOA-level aggregation
Force-level enrichment
LSOA population and deprivation join
Add force-level population and deprivation
ONS LSOA to ward lookup
Ward names and geographical features
Final validation
Grain, nulls, joins and reconciled counts
One BI-ready reporting CSV
Power BI reporting: Phase 2